Bogleheads

Approximating total stock market

*This article contains details **specific to United States (US) investors**. If you are a non-US investor, **do not use** any funds or ETFs listed here. They expose you to harmful US dividend tax and estate tax risks. Proceed **only** if you have verified these risks do not apply to you.*

Approximating total stock market shows how you can combine funds to mimic the composition of the U.S. stock market.

Investors who want to invest in a Total Stock Market Index Fund often face a situation where they have to approximate it with the funds available in their employer sponsored plans, perhaps also using funds in other accounts, for example a Roth IRA.

This page shows some examples of how to approximate a Total Stock Market Index Fund with funds that cover specific parts of the market.

Total stock market index

This table shows Total Stock Market Index Funds from four different providers. The aim is to approximate these Morningstar style boxes using available funds chosen from the two-fund and three-fund tables below.

Total Stock Market index funds	Morningstar style box
100% Vanguard Total Stock Market Index Fund (VTSAX) (https://investor.vanguard.com/mutual-funds/profile/VTSAX) xray (https://lt.morningstar.com/3y3wd9echv/xray/default.aspx?LanguageId=en-GB&PortfolioType=3&SecurityTokenList=FOUSA00L83%5D2%5D0%5DFOALL$$ALL&values=100&CurrencyId=USD&from=edit holding) -or- 100% Fidelity ® Total Market Index Fund (FSKAX) (https://nb.fidelity.com/public/workplacefunds/summary/2361) Updated 30 June 2024	Value / Blend / Growth Large: 12 / 25 / 34 Mid: 6 / 9 / 5 Small: 3 / 3 / 2
100% iShares Core S&P Total U.S. Stock Market ETF (ITOT) (https://www.ishares.com/us/products/239724/ishares-core-sp-total-us-stock-market-etf) Updated 23 July 2021	Value / Blend / Growth Large: 16 / 24 / 32 Mid: 6 / 9 / 5 Small: 3 / 4 / 2
100% Schwab U.S. Broad Market ETF (SCHB) (https://www.schwabassetmanagement.com/products/schb) Updated 23 July 2021	Value / Blend / Growth Large: 17 / 24 / 33 Mid: 6 / 9 / 5 Small: 3 / 3 / 2
100% Fidelity ZERO Total Market (FZROX) (https://fundresearch.fidelity.com/mutual-funds/summary/31635T708) Updated 23 July 2021	Value / Blend / Growth Large: 17 / 25 / 30 Mid: 6 / 9 / 5 Small: 3 / 3 / 2

Approximating total stock with two funds

These examples match the composition of the total US stock market using two funds.

Match Total Stock Index with two funds	Morningstar style box
87.5% Vanguard 500 Index Fund (VFIAX) (https://investor.vanguard.com/mutual-funds/profile/VFIAX) 12.5% Vanguard Extended Market Index Fund (VEXAX) (https://investor.vanguard.com/mutual-funds/profile/VEXAX) Updated 31 January 2026[1]	Value / Blend / Growth Large: 12 / 24 / 34 Mid: 6 / 9 / 6 Small: 3 / 4 / 3
85% Vanguard Large Cap Index Fund (VLCAX) (https://investor.vanguard.com/mutual-funds/profile/VLCAX) 15% Vanguard Small-Cap Index Fund (VSMAX) (https://investor.vanguard.com/mutual-funds/profile/VSMAX) Updated 31 May 2021	Value / Blend / Growth Large: 17 / 26 / 29 Mid: 6 / 9 / 5 Small: 3 / 4 / 2
92% iShares Russell 1000 ETF (IWB) (https://www.ishares.com/us/products/239707/ishares-russell-1000-etf) 8% iShares Russell 2000 ETF (IWM) (https://www.ishares.com/us/products/239710/ishares-russell-2000-etf)[note 1] Updated 31 May 2021	Value / Blend / Growth Large: 17 / 26 / 28 Mid: 6 / 9 / 4 Small: 3 / 4 / 2
85% Vanguard 500 Index Fund (VFIAX) (https://investor.vanguard.com/mutual-funds/profile/VFIAX) 15% Vanguard Small-Cap Index Fund (VSMAX) (https://investor.vanguard.com/mutual-funds/profile/VSMAX) Updated 31 May 2021	Value / Blend / Growth Large: 18 / 27 / 28 Mid: 6 / 9 / 4 Small: 3 / 4 / 2

90% Vanguard 500 Index Fund (VFIAX) (https://investor.vanguard.com/mutual-funds/profile/VFIAX) 10% iShares Russell 2000 Index (IWM) (https://www.ishares.com/us/products/239710/ishares-russell-2000-etf) Updated 31 May 2021	**Value**	**Blend**	**Growth**	
	19	29	29	**Large**
	5	7	2	**Mid**
	3	4	2	**Small**
90% Vanguard 500 Index Fund (VFIAX) (https://investor.vanguard.com/mutual-funds/profile/VFIAX) 10% Vanguard Tax-Managed Small-Cap Fund Inv (VTMSX) (https://investor.vanguard.com/investment-products/mutual-funds/profile/vtmsx) Updated 31 May 2021	**Value**	**Blend**	**Growth**	
	19	29	29	**Large**
	5	7	2	**Mid**
	3	5	2	**Small**
83% Fidelity ZERO Large Cap Index Fund (FNILX) (https://fundresearch.fidelity.com/mutual-funds/summary/315911628) 17% Fidelity ZERO Extended Market Fund (FZIPX) (https://fundresearch.fidelity.com/mutual-funds/summary/315911636) Updated 23 July 2021	**Value**	**Blend**	**Growth**	
	17	25	30	**Large**
	6	9	5	**Mid**
	3	4	2	**Small**

Approximating total stock with three funds

These examples match the composition of the total US stock market using three funds.

Match Total Stock Index with three funds	Morningstar style box
85% Vanguard 500 Index Fund (VFIAX) (https://investor.vanguard.com/mutual-funds/profile/VFIAX) 10% iShares S&P MidCap 400 ETF (IJH) (https://www.ishares.com/us/products/239763/ishares-core-sp-midcap-etf) 5% iShares S&P SmallCap 600 ETF (IJR) (https://www.ishares.com/us/products/239774/ishares-core-sp-smallcap-etf) Updated 31 May 2021	Value / Blend / Growth Large: 18 / 27 / 28 Mid: 6 / 9 / 3 Small: 3 / 5 / 1
83% Fidelity 500 Index Fund (FXAIX) (https://fundresearch.fidelity.com/mutual-funds/analysis/315911750) 8% Fidelity Mid Cap Index Fund (FSMDX) (https://fundresearch.fidelity.com/mutual-funds/summary/316146265) 9% Fidelity Small Cap Index Fund (FSSNX) (https://fundresearch.fidelity.com/mutual-funds/summary/316146182) Updated 31 May 2021	Value / Blend / Growth Large: 18 / 28 / 26 Mid: 6 / 9 / 4 Small: 2 / 4 / 2
85% Vanguard S&P 500 ETF (VOO) 10% Vanguard S&P Mid-Cap 400 ETF (IVOO) 5% Vanguard Russell 2000 ETF (VTWO)[note 2] Updated 22 July 2021	Value / Blend / Growth Large: 17 / 24 / 31 Mid: 6 / 9 / 3 Small: 3 / 5 / 2

Table notes

For the above tables:

- The style box numbers were computed by Morningstar *Instant X-Ray* by entering each fund ticker as a holding, and entering the dollar value based on the percentage; for example, for VFIAX at 81% and VEXAX at 19%, enter $81 and $19 respectively and observe the computed Morningstar Style Box on the Instant X-Ray tab. Enter different dollar amounts (always adding up to $100) in a trial and error fashion until you replicate the same computation as Total Stock Market, or as close as you can get.[note 3]
- Computation used the method of least squares (http://en.wikipedia.org/wiki/Least_squares)
- If you are thinking of using an S&P MidCap 400 fund, for example, you might consider the example above that uses IJH. However, there may be slight implementation differences between the S&P MidCap 400 fund you have access to and IJH.

- Two funds may appear to track the same portion of the market (say, mid-cap), but their holdings may differ depending on the indexes they track.

Stock market classifications

Style box format

Style boxes use the following format. Numbers shown are percentages. Because values are rounded, the percentages may not total exactly 100%.

Value	Blend	Growth	
LV	LB	LG	**Large**
MV	MB	MG	**Mid**
SV	SB	SG	**Small**

Size

Stocks may be classified by the size of the corporation. This is most commonly done by market capitalization. Market capitalization is simply a measurement found by taking a stock's current share price and multiplying it by the number of stock shares outstanding.

Exact market cap ranges will vary among different financial and rating institutions, but there are three different terms commonly used to describe stocks by their general size.

- **Large cap stocks** have a market cap over $10 billion.
- **Mid cap stocks** have a market cap between $2 billion and $10 billion.
- **Small cap stocks** have a market cap between $300 million and $2 billion.

The Morningstar style box uses percentages rather than fixed dollar thresholds for size classification.[2]

Style

Stocks may also be classified by "style," one of **Value**, **Blend**, or **Growth**.

- **Growth stocks** are companies that are growing their profits at a very fast rate and are expected to continue to grow at an increasing rate.
- **Value stocks** are stocks that tend to trade at deep discount relative to their intrinsic value (as defined by profits, book value, and so on).

Investors commonly think of growth stocks as "high flying companies" and value stocks as "distressed companies." Both growth and value stocks have taken turns leading and lagging one another during different markets and economic conditions.[note 4]

Notes

1. See this Bogleheads forum post: "Re: Follow up questions - total stock market approximation" (https://www.bogleheads.org/forum/viewtopic.php?p=1571134#p1571134), hoppy08520. Jan 03, 2013 for more on combining these two funds to approximate total stock market.
2. This combination might interest you if you have a 401(k) plan held at John Hancock Funds, which offers 3 similar funds: JEINX tracks the S&P 500, JECIX follows the S&P Midcap 400, and JESIX tracks the Russell 2000. (As of November 1, 2015, the John Hancock ticker symbols are no longer available in Morningstar Instant X-Ray.) In addition, this also similarly mirrors the offerings available to State of CA employees who invest at SavingsPlusNow, which offers Large Cap, Mid Cap, and Small Cap index funds that mirror the same. (As of March 16, 2023)
3. Morningstar's *Instant X-Ray* portfolio analysis tool has been discontinued effective April 17, 2025. Source: "Upcoming changes to our membership offerings, tools, and features" (https://global.morningstar.com/en-gb/updates/upcoming-changes-our-membership-offerings-tools-features). *Morningstar*. April 15, 2025. Retrieved March 10, 2026. Forum member LadyGeek has developed an alternative Equity style box calculator (https://docs.google.com/spreadsheets/d/113HpxXv5NrKWDXpv_xGNIVeOQk3giuRD/edit?pli=1&gid=63194195#gid=63194195) spreadsheet. Discussion and support is in Bogleheads forum topic: "Wiki - Morningstar's free Instant X-ray tool no longer available - Alternatives?" (https://www.bogleheads.org/forum/viewtopic.php?t=469099), LadyGeek. March 08, 2026.
4. You can see the alternating performance of the growth and value stocks on the Callan periodic table of investment returns, which visualizes performance of different asset classes.

See also

- Replicating total stock market with the Thrift Savings Plan
- Approximating total international stock market

References

1. Bogleheads forum topic: "Replicating total stock market with s&p 500 and extended market index" (https://www.bogleheads.org/forum/viewtopic.php?t=467459). January 31, 2026
2. "Fact sheet: The new Morningstar style box methodology" (https://web.archive.org/web/20220822012855/http://news.morningstar.com/pdfs/FactSheet_StyleBox_Final.pdf) (PDF). Morningstar. Archived from the original (http://news.morningstar.com/pdfs/FactSheet_StyleBox_Final.pdf) (PDF) on August 22, 2022.

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